

Coffee Shop Sales Analytics Dashboard

End-to-End SQL S Power BI Business Intelligence Project

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Executive Summary

This project analyzes coffee shop sales using MySQL and Power BI to build an interactive dashboard with KPIs, Month-over-Month growth, product, store and hourly sales analysis.

Project Snapshot

Item	Details
Tools	MySQL, Power BI, DAX
Dataset	Coffee Shop Sales
KPIs	Sales, Orders, Quantity, MoM Growth
Fact Table	Transactions
Dimension	Date Table
Special Features	Custom Tooltips, Heatmap, Dynamic Labels

Business Problem

The coffee shop operates across multiple store locations and records thousands of sales transactions every day. Although the sales data is available, management lacks a centralized dashboard to monitor business performance, identify trends, and make data-driven decisions efficiently.

Project Objective

Develop a single interactive Power BI dashboard to monitor sales performance, identify top-performing products and stores, analyze peak business hours, and support faster business decision-making.

KPI Requirements

1. Total Sales Analysis

- Calculate the total sales for each respective month.
- Determine the month-on-month increase or decrease in sales.
- Calculate the difference in sales between the selected month and the previous month.

Solution

- Created a **Total Sales KPI Card** to display the current month's sales.
 - Developed **CM Sales** and **PM Sales** DAX measures using Time Intelligence.
 - Implemented **Month-over-Month (MoM) Growth** and **Sales Difference** calculations to compare the selected month with the previous month.
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2. Total Orders Analysis

- Calculate the total number of orders for each respective month.
- Determine the month-on-month increase or decrease in the number of orders.
- Calculate the difference in the total number of orders between the selected month and the previous month.

Solution

- Created a **Total Orders KPI Card**.
 - Used **DISTINCTCOUNT ()** to calculate unique orders.
 - Built **CM Orders**, **PM Orders**, and **MoM Order Growth** measures to analyze monthly performance.
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3. Total Quantity Sold Analysis

- Calculate the total quantity sold for each respective month.
- Determine the month-on-month increase or decrease in the total quantity sold.
- Calculate the difference in the total quantity sold between the selected month and the previous month.

Solution

- Created a **Total Quantity KPI Card**.
 - Used **SUM()** to calculate the total quantity sold.
 - Implemented **CM Quantity**, **PM Quantity**, and **MoM Quantity Growth** measures.
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Chart Requirements

1. Calendar Heat Map

- Implement a calendar heat map that dynamically adjusts based on the selected month from a slicer.
- Each day in the calendar should be color-coded to represent sales volume, with darker shades indicating higher sales.
- Implement tooltips to display detailed metrics (Sales, Orders, Quantity) when hovering over a specific day.

Solution

- Built a **dynamic Calendar Heat Map** using the Date Table.
 - Added a **Month slicer** for dynamic filtering.
 - Applied conditional formatting to highlight high- and low-sales days.
 - Created a **custom tooltip page** showing Sales, Orders, and Quantity for the selected date.
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2. Sales Analysis by Weekdays and Weekends

- Segment sales data into weekdays and weekends to analyze performance variations.
- Provide insights into whether sales patterns differ significantly between weekdays and weekends.

Solution

- Created a **Weekday/Weekend classification** in the Date Table using DAX.
 - Built a **donut chart** to compare sales contribution.
 - Enabled dynamic filtering based on the selected month.
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3. Sales Analysis by Store Location

- Visualize sales data by different store locations.
- Include month-over-month (MoM) difference metrics based on the selected month in the slicer.
- Highlight MoM sales increase or decrease for each store location to identify trends.

Solution

- Developed a **Store Performance Bar Chart**.
 - Added **dynamic labels** to show total sales for each location.
 - Displayed **MoM growth indicators** for each store.
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4. Daily Sales Analysis with Average Line

- Display daily sales for the selected month using a line chart.
- Incorporate an average line on the chart to represent the average daily sales.
- Highlight bars exceeding or falling below the average sales to identify exceptional sales days.

Solution

- Created a **Daily Sales Trend Chart**.
 - Calculated **Daily Average Sales** using DAX.
 - Added an **Average Line** to compare daily performance.
 - Used data labels to highlight sales values.
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5. Sales Analysis by Product Category

- Analyze sales performance across different product categories.
- Provide insights into which product categories contribute the most to overall sales.

Solution

- Created a **Product Category Bar Chart**.
 - Added dynamic labels to display category names and sales values.
 - Sorted categories by sales to identify top contributors.
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6. Top 10 Products by Sales

- Identify and display the top 10 products based on sales volume.
- Allow users to quickly visualize the best-performing products.

Solution

- Built a **Top Products Bar Chart**.
 - Ranked products based on total sales.
 - Displayed product names and corresponding sales values.
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7. Sales Analysis by Days and Hours

- Utilize a heat map to visualize sales patterns by days and hours.
- Implement tooltips to display detailed metrics (Sales, Orders, Quantity) when hovering over a specific day-hour combination.

Solution

- Created a **Day & Hour Heat Map**.
- Extracted the **Hour** from transaction time using DAX.
- Designed a **custom tooltip page** to display Sales, Orders, and Quantity when hovering over a specific day-hour combination.

Solution Workflow

CSV → MySQL → SQL Analysis → Power BI → DAX → Interactive Dashboard

Dashboard Walkthrough

This section explains the design of the Power BI dashboard and how each visual helps business users monitor sales performance and make data-driven decisions.

Dashboard Overview

The dashboard is designed as a one-page executive report with interactive filters, KPI cards, trend charts, category analysis, store performance analysis and hourly sales heatmap. It enables managers to quickly identify business performance without navigating multiple reports.

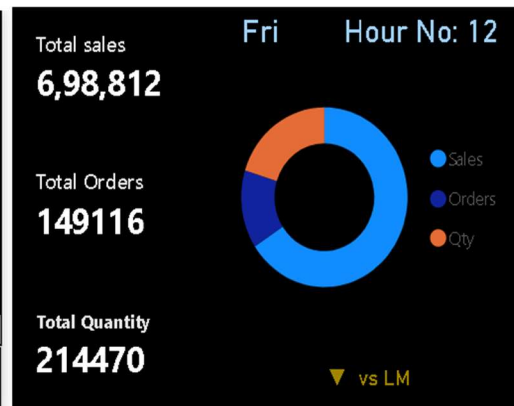
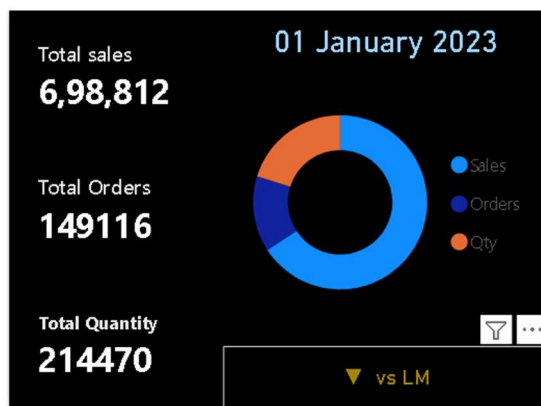
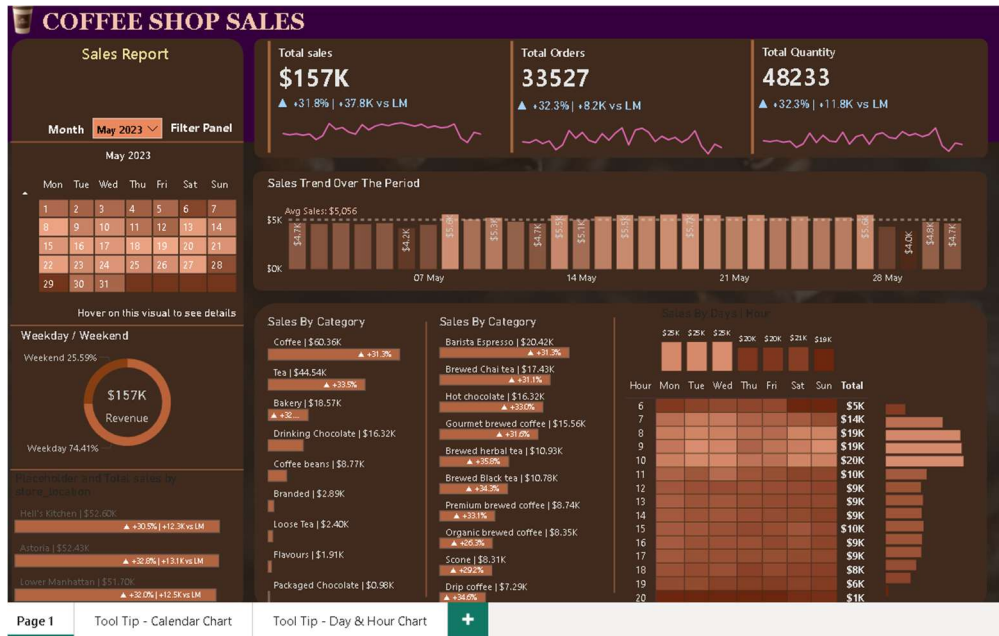
Dashboard Component	Purpose	Business Value
Month Slicer	Filter dashboard by selected month	Analyze monthly performance
KPI Cards	Show Total Sales, Orders and Quantity with MoM comparison	Quick executive summary
Sales Trend	Display daily sales trend	Identify high and low performing days
Category Analysis	Compare product category sales	Find major revenue contributors
Product Analysis	Compare product type performance	Support inventory planning
Store Performance	Compare store locations	Identify best performing stores
Weekday vs Weekend	Compare revenue distribution	Understand customer buying pattern
Day S Hour Heatmap	Visualize peak hours by weekday	Support staffing and operations
Custom Tooltips	Show detailed metrics on hover	Reduce dashboard clutter while keeping insights available

Dashboard Design Decisions

- Coffee-inspired brown color palette was selected to match the business domain.
- Interactive slicers allow users to analyze different months without changing the report.
- Sparkline KPIs provide instant trend visibility.
- Custom tooltip pages improve user experience without overcrowding the dashboard.
- Heatmap was selected instead of a standard chart to identify peak business hours more effectively.

Key Features Implemented

- Dynamic Month Selection
- Month-over-Month KPI Comparison
- Dynamic Labels using DAX
- Calendar Tooltip
- Day S Hour Tooltip
- Interactive Filtering
- Executive Dashboard Layout



Technical Implementation

This section explains the technical implementation of the Coffee Shop Sales Analytics Dashboard, including the data architecture, SQL workflow, data model, and DAX measures used to build the solution.

1. Project Workflow

CSV Dataset

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MySQL Database

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Data Cleaning & Validation using SQL

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Business Analysis using SQL Queries

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Power BI Data Model

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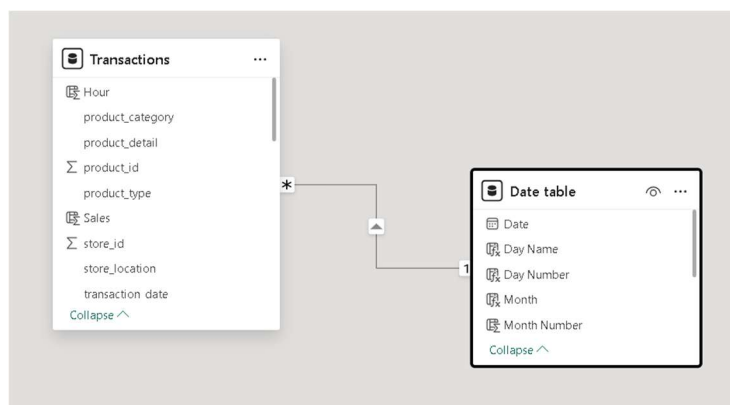
DAX Measures

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Interactive Dashboard

2. Data Model

The project follows a Star Schema data model. The Transactions table acts as the Fact Table, while the Date Table acts as the Dimension Table. A One-to-Many relationship is created between Date Table[Date] and Transactions[transaction_date], which enables efficient filtering and supports Time Intelligence calculations.



3. SQL Implementation

Task	Purpose
Import Dataset	Load raw sales data into MySQL
Data Validation	Check missing and incorrect values
Data Cleaning	Prepare clean dataset for reporting
Aggregation	Calculate sales, quantity and orders
Business Analysis	Generate insights before visualization

4. DAX Measures

Measure	Purpose	Business Use
Total Sales	Calculate overall revenue	Primary KPI
Total Orders	Count unique transactions	Customer demand
Total Quantity	Total quantity sold	Sales volume
CM Sales / Orders / Quantity	Current month values	Current performance
PM Sales / Orders / Quantity	Previous month values	Historical comparison
MoM Growth	Month-over-Month comparison	Growth tracking
Daily Avg Sales	Average daily revenue	Performance benchmarking
Dynamic Labels	Display value with category/store	Better readability
Hour	Extract hour from transaction time	Hourly analysis

5. Key Technical Highlights

- Star Schema Data Model
- Time Intelligence using TOTALMTD and DATEADD
- Variables (VAR) for optimized DAX
- Dynamic KPI Cards
- Custom Labels using SELECTEDVALUE
- Interactive Filtering with Date Table

Business Insights & Recommendations

This section summarizes the key business findings obtained from the dashboard and provides practical recommendations that management can use to improve business performance.

1. Business Insights

Insight Area	Observation
Overall Sales Performance	The KPI cards provide an instant overview of Total Sales, Total Orders and Total Quantity Sold, allowing management to monitor overall business health.
Monthly Performance	Month-over-Month KPIs help identify whether sales performance is improving or declining compared to the previous month.
Store Performance	Store comparison visuals highlight the highest and lowest performing locations, enabling location-wise performance evaluation.
Product Category Analysis	Category-wise analysis shows which product categories contribute the most to revenue.
Product Performance	Product-level analysis identifies best-selling and low-selling products for inventory planning.
Peak Business Hours	The Day S Hour heatmap reveals customer traffic patterns and peak operating hours.
Weekday vs Weekend	Comparing weekdays and weekends helps understand customer buying behaviour and demand fluctuations.

2. Business Recommendations

- Maintain higher inventory levels for high-selling products.
- Schedule more staff during peak business hours identified by the heatmap.
- Launch promotional offers for slow-moving products.
- Analyze the operating practices of top-performing stores and apply them to lower-performing stores.
- Track Month-over-Month KPIs regularly to measure business growth.
- Use the dashboard during weekly and monthly review meetings for faster decision-making.

3. Business Value Delivered

- Centralized reporting in a single dashboard.
- Faster access to business KPIs.
- Improved visibility into store and product performance.
- Better operational planning using hourly sales analysis.
- Support for data-driven decision-making.

Project Conclusion & Future Scope

Project Conclusion

This project successfully transformed raw coffee shop transaction data into an interactive Business Intelligence dashboard using MySQL and Power BI. The dashboard enables business users to monitor KPIs, analyze sales trends, evaluate store and product performance, and make informed decisions using data.

Key Achievements

- Developed an end-to-end SQL and Power BI analytics solution.
- Created dynamic KPI cards for Sales, Orders and Quantity.
- Implemented Month-over-Month growth analysis using DAX.
- Designed interactive dashboard with slicers and custom tooltips.
- Applied Star Schema for efficient data modeling.
- Built business-focused visualizations for management reporting.

Skills Demonstrated

Technical Skills	Business Skills
MySQL S SQL	Business Problem Solving
Power BI	Decision Support
DAX	KPI Development
Data Modeling	Business Analysis
Data Visualization	Performance Monitoring
Dashboard Design	Insight Generation

Future Scope

- Add Profit and Margin Analysis.
- Include Customer Segmentation.
- Connect to Live SQL Database.
- Implement Scheduled Data Refresh.
- Add Sales Forecasting.
- Implement Row-Level Security (RLS).

Final Project Summary

This project demonstrates practical knowledge of SQL, Data Modeling, DAX and Power BI by solving a real-world retail sales analytics problem. It reflects both technical implementation skills and the ability to convert data into meaningful business insights.